

# Khushal Midha

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## EDUCATION

Indian Institute of Information Technology, Lucknow  
B.Tech in Computer Science and Artificial Intelligence – CGPA: 8.72/10

Courses: OOPS, DBMS, OS, DSA  
August 2023 – July 2027

## EXPERIENCE

SDE Intern – Coffeee.io

Remote — Jun 2025 – Aug 2025

- Built a **project-based assessment module** (Java, Spring Boot, React, PostgreSQL) where candidates debug live codebases via **error logs**, graded automatically against hidden test cases.
- Developed a **read-only AI assistant** (LLMs, Redis, Docker) to provide contextual hints without edit access, preserving evaluation integrity; validated in **staging**.

## PROJECTS

- **1. MediPulse**  Feb 2025 – Present  
*MERN, FastAPI, Hugging Face Transformers, Redis, Kafka, Socket.IO, WebRTC, RAG* Live Demo 
  1. Built a **multi-tenant healthcare SaaS** with **125 REST APIs**, **24 MongoDB models**, and **34 React pages**, supporting **appointments**, **live OPD queues**, **WebRTC consultations**, and **hospital portals** with **7-role RBAC** via **JWT & Google OAuth**.
  2. Developed an **AI triage engine** combining a **Gemini agent**, **3 Hugging Face models**, and a custom **RAG pipeline** (using **text-embedding-004 & Cosine Similarity**) via **FastAPI** for accurate **disease prediction**, **specialty routing**, and **context-aware patient memory**.
  3. Engineered **real-time OPD queues** and **virtual wallets** using **Socket.IO**, **Redis**, and **Kafka**, implementing **atomic transactions**, **distributed locks**, **idempotent refunds**, and **asynchronous event processing** for **concurrency-safe operations**.
- **2. TriageAgent**  May 2026 - Present  
*Python, LiteLLM, FAISS, RAG, Sentence-Transformers, LLMs*
  1. Architected a provider-agnostic **AI support agent** utilizing **LiteLLM** and **FAISS** for semantic retrieval (RAG), improving intent classification accuracy by **37%** (achieving 81.4%) over keyword baselines.
  2. Engineered a **6-signal risk escalation engine** and an automated **LLM-as-a-Judge** evaluation pipeline, achieving a perfect **1.0 Cohen's Kappa** agreement score and eliminating manual QA bottlenecks.
- **3. ThreatGraph**  Jan 2026 - Present  
*Python, PyTorch, FastAPI, Kafka, PostgreSQL, NetworkX, FAISS, RAG, GNNs*
  1. Architected a **spatial-temporal threat detection engine** fusing **Graph Neural Networks** and **Transformers**, processing **3,000+ events/sec** via **Kafka**, achieving **92% PR-AUC** on imbalanced attacks.
  2. Built a full-stack **automated incident response platform** integrating a **RAG-powered LLM analyst** to map attack paths to **MITRE ATT&CK** and trigger real-time endpoint quarantines via **FastAPI**.

## ACHIEVEMENTS

1. Rated as **Expert** Codeforces(1809), **5-star** CodeChef (2137), **Guardian** LeetCode (2137).
2. Solved **2500+ DSA problems** across Codeforces and LeetCode strengthening problem-solving skills.
3. Selected for **Amazon ML School 2025**, among top 3000 from 85,000 applicants Proof 
4. Achived a rank of **662 in ICPC Amritapuri Region 2024-25** by the team name **Overshadowed**. Ranklist 
5. Secured rank of **186 in Facebook Meta Hackercup Round-3** among 14000+ Participants Ranklist 
6. Qualified **Adobe Gensolve Hackathon**, ranked **top 5%** among 54000+ participants. Certificate 
7. Topped among **250+ peers** in **Freshers Cup 2024** by CP Wing, Axios, IIIT Lucknow
8. Achieved All India Rank **15,503 in JEE Advanced 2023** and Rank **9,932 in JEE Mains 2023**.
9. Secured rank **996 in Goldman Sachs Hackathon** under the username **midhakhushal5** in the CS round.

## SKILLS & OTHERS

1. **Languages:** C, C++, Python, JavaScript, Java, SQL
2. **Frontend:** React.js, TypeScript, HTML, CSS, Tailwind CSS, Bootstrap
3. **Backend:** Node.js, Express.js, Spring Boot, Redis, Kafka, Fast API
4. **Database:** MongoDB, MySQL, PostgreSQL
5. **Machine Learning & AI:** PyTorch, XGBoost, Scikit-learn, Transformers, MLOps (MLflow), LLMs, Agentic AI, Prompt Engineering
6. **Courses:** Object Oriented Programming, Database Managment System, Computer Networks, Operating System
7. **Technical Skills:** Data Structures Algorithms, Competitive Programming, Web Development, REST API Design